STAT 8305: Pattern Recognition
Homework 2: Cross-Validation for Dependent Data

Central goal. The purpose of this homework is to understand how dependence among observations affects cross-validation (CV). You will describe a dependence structure, design an appropriate training and validation strategy, and formulate a CV procedure for evaluating prediction performance.

Background: Cross-Validation for Dependent Data

Cross-validation is widely used to evaluate prediction performance and select statistical models. In standard K-fold CV, observations are randomly divided into K subsets. Each subset is used for validation, while the remaining observations are used for model fitting.

However, observations are not always independent. For example, measurements collected over time, observations from nearby spatial locations, and repeated measurements from the same individual may exhibit substantial dependence.

When training and validation observations are dependent, random splitting may lead to misleading estimates of prediction error. An appropriate CV procedure should therefore account for the dependence structure and the intended prediction problem.

Dependence can be described through statistical assumptions or parameterized correlation models. For example, for temporally ordered random errors, one may consider

$$\mathrm{Corr}(\varepsilon_i, \varepsilon_j) = \rho^{|i-j|}, \qquad |\rho| < 1,$$

where ρ controls the strength of temporal dependence. Other dependence structures may require different assumptions or additional parameters.

Importantly, accounting for dependence does not necessarily require complete independence between training and validation observations. The objective is to construct a validation procedure that avoids inappropriate information leakage and provides a meaningful evaluation of prediction performance.

Homework Question: Cross-Validation for Dependent Data

Suppose we observe a dataset

$$\mathcal{D}_n = \{(X_i, Y_i) : i = 1, \ldots, n\},$$

where X_i and Y_i denote the input and output, respectively.

Assume that a prediction model is available and can be fitted using all or part of the data. Once fitted, the model produces predicted values $\widehat{Y}_i$ for the corresponding inputs X_i.

Question. Design a cross-validation procedure for evaluating prediction performance when observations are dependent. You may consider temporal, spatial, grouped, or another appropriate dependence structure.

(a) **Dependence assumptions.** Describe a possible dependence structure among the observations. Clearly state your assumptions and, if appropriate, specify a correlation model with sufficient parameters to characterize the dependence.

(b) **Data splitting.** Explain how you would divide the data into training and validation sets. Clearly define which observations are used for model fitting and which are used for evaluating predictions.

(c) **Cross-validation formula.** Define an appropriate prediction loss function $L(Y_i, \widehat{Y}_i)$ and write down the mathematical formula for the CV error under your proposed splitting procedure. Clearly define all notation used in the formula.

(d) **Justification.** Explain why your splitting strategy is appropriate under your dependence assumptions. In particular, discuss how it avoids information leakage, reduces problematic dependence between training and validation observations when necessary, and provides a meaningful evaluation of prediction performance.

Submission

Submit a short report addressing parts (a)–(d). Clearly state your dependence assumptions, define the training and validation sets, and provide the mathematical formula for your proposed CV procedure. No programming or model fitting is required. The purpose is to practice designing and justifying a statistically appropriate cross-validation procedure for dependent observations.